

Bitwise AND operators:

1. $13 \& 11$

13	11	13 & 11
1	1	1
1	0	0
0	1	0
1	1	1
2^4 16	2^3 8	2^2 4
2^1 2	2^0 1	
1 0 0 1 $\Rightarrow 9$		

2. $14 \& 9$

14	9	14 & 9
1	1	1
1	0	0
1	0	0
0	1	0
2^4 16	2^3 8	2^2 4
2^1 2	2^0 1	
1 0 0 0 $\Rightarrow 8$		

3. $12 \& 7$

12	7	12 & 7
1	0	0
1	1	1
0	1	0
0	1	0
2^3 8	2^2 4	2^1 2
2^0 1		
0 1 0 0 $\Rightarrow 4$		

Bitwise OR operators:

1. $12 \mid 13$

12	13	12 13
1	1	1
1	1	1
0	0	0
0	1	1
2^3 8	2^2 4	2^1 2
2^0 1		
1 1 0 1 $\Rightarrow 13$		

2. $16 \mid 17$

16	17	16 17
1	1	1
0	0	0
0	0	0
0	0	0
0	1	1
2^4 16	2^3 8	2^2 4
2^1 2	2^0 1	
1 0 0 0 1 $\Rightarrow 17$		

3. $20 \mid 25$

20	25	20 25
1	1	1
0	1	1
1	0	1
0	0	0
0	1	1
2^4 16	2^3 8	2^2 4
2^1 2	2^0 1	
1 1 1 0 1 $\Rightarrow 29$		

Bitwise XOR operators:

1. $22 \wedge 12$

22	12	22 ^ 12
1	0	1
0	1	1
1	1	0
1	0	1
0	0	0
2^4 16	2^3 8	2^2 4
2^1 2	2^0 1	
1 1 0 1 0 $\Rightarrow 26$		

2. $8 \wedge 16$

8	16	8 ^ 16
0	1	1
1	0	1
0	0	0
0	0	0
0	0	0
2^4 16	2^3 8	2^2 4
2^1 2	2^0 1	
1 1 0 0 0 $\Rightarrow 24$		

3. $10 \wedge 30$

10	30	10 ^ 30
0	1	1
1	1	0
0	1	1
1	1	0
0	0	0
2^4 16	2^3 8	2^2 4
2^1 2	2^0 1	
1 0 1 0 0 $\Rightarrow 20$		